SM2501/FSG3084 Biomechanics of human movement, Autumn 2025

Wk	Date	Time	Room	Topics	Reading	Assignment due
44	Mon 27-Oct	10-12	U21	L1: Course contents Anatomy, Motion definitions, biological tissues	Ch. 1	
	Tues 28-Oct	10-12	U21	L2: Soft tissues biomechanics, Muscle tissue and activation	Ch. 4-5	
	Wed 29-Oct	15-17	U21	L3: Motion and motion analysis Intro to OpenSim comp. labs	Ch. 7	
	Thurs 30-Oct	9.15 10.15 11.15	MoveAbility Lab, OB18	*Experiments* in Promobilia MoveAbility Lab, Osquars Backe 18, 4th fl (compulsory)		
45	Mon 3-Nov	10-12	Christopher	*Computer Lab 1*: Tutorial 1		*Canvas quiz*
	Tues 4-Nov	10-12	U21	L4: Inverse dynamics	Ch. 8	
	Wed 5-Nov	14-16 *16-18*	U21 *MoveAbility*	L5: Muscle biomechanics and physiology I (R Wang) *TA Office Hours*	Ch. 5-6	
46	Mon 10-Nov	10-12	U21	L6: Muscle biomechanics II	Ch. 5-6	
	Tues 11-Nov	10-12 *13-15*	U21 *MoveAbility*	L7: Kinematics of normal walking *TA Office Hours (J Maier)*	Ch. 2	
	Thurs 13-Nov	13-15 15-17	U21 Christopher/ zoom	L8: Kinetics and muscle activities in walking *Computer Lab 2*: OpenSIM IK + ID	Ch. 11	 *Canvas quiz*
	Fri 14-Nov					*Proj1 (Motion biomechanics) due 12.00*
47	Mon 17-Nov	10-12	Christopher/ zoom	*Computer Lab 3*: OpenSIM RRA + Static Optimization	Ch. 9-10	*Canvas quiz*
	Tues 18-Nov	10-12 *13-15*	U21 *MoveAbility*	L9: Muscles III *TA Office Hours (J Maier)*		
	Thur 20-Nov	13-15	U21	L10: Motor control and muscle modelling (R Wang)		
48	Mon 24-Nov	8-10	Q2	L11: Pathological walking I	Ch. 3	
	Tues 25-Nov	10-12 *13-15*	U21 *MoveAbility*	L12: Pathological walking II *TA Office Hours (J Maier)*		
	Thurs 27-Nov	13-15	U21	L13: Optimization in biomechanics + shoulder modelling (J Maier)	Ch. 9	*Proj2 (OpenSIM IK+ID +RRA+SO) due 12.00*
49	Mon 1-Dec	10-12	U21	L14: Biomechanics and orthopedic surgery (Guest S Gantelius)	Ch. 3	
	Thurs 4-Dec	8-10 *10-12*	U31 *MoveAbility*	L15: Biomechanics in sports (Guest T Arndt) *TA Office Hours (J Maier)*	Ch. 13	
50	Mon 8-Dec	10-12	U21	L16: Weakness compensation, performance enhancement, walking aids		*Proj3 (OpenSIM, exoskeleton) due 9.30*
	Thurs 11-Dec	15-17	U21	L17: Biomechanics research, summary		
2-3	TBD	TBD	TBD	Final project presentations (in time slots). Date and time announced later		1-page summary, due date TBD

Course responsible/teacher: Professor Lanie Gutierrez-Farewik (lanie@kth.se).
Asst teacher: Assoc. Prof. Ruoli Wang (ruoli@kth.se)
Course assistant: Jennifer Maier (jenmai@kth.se)
Guest lecturers: Stefan Gantelius (MD, PhD, Head Dept. Orthopaedics, Danderyd Hosp.), Toni Arndt (Professor, GIH)
Course book: Biomechanics of Movement; The Science of Sports, Robotics, & Rehabilitation, TK Uchida & SL Delp
Computer labs will be at Christopher and at https://kth-se.zoom.us/j/7600710202. The first computer lab will only be IRL at Christopher. The second and third will be hybrid; you can attend IRL or via zoom.
If discrepancies between this schedule and the KTH/ Canvas schedule occur, always defer to the Canvas schedule.